Nimbus Harbor

Platform Operations Handbook

Release 4.2.0 "Kestrel" | Document ID NHB-OPS-0042

Fictional document created for retrieval-augmented generation (RAG) benchmarking.

1. Platform Overview

Nimbus Harbor is an internal developer platform that lets product teams build, ship and operate services without managing infrastructure directly. Release 4.2.0, codenamed Kestrel, shipped on 18 March 2025. The platform is made up of four core services.

Service	Role	Default port
Lighthouse	API gateway and request router	8472
Ballast	Asynchronous job queue	9115
Tidewater	Metrics pipeline	9302
Driftnet	Log aggregator	5514

1.1 Regional clusters

Nimbus Harbor runs in three regional clusters: **eu-north-quartz**, **ap-south-onyx** and **us-east-amber**. The control plane lives in eu-north-quartz. If the control plane is unreachable for more than 120 seconds, a failover controller named Pilotage promotes us-east-amber to primary. The ap-south-onyx cluster never hosts the control plane and only serves workloads for the Asia-Pacific region.

2. API and Authentication

Every request to Lighthouse must carry a bearer token. User tokens begin with the prefix **hbt**_ and expire after 55 minutes. Service accounts use tokens that begin with **hsa**_ and remain valid for 24 hours. A token is renewed by sending `POST /v2/tokens/renew`. Tokens are signed with the EdDSA algorithm, and the signing keys are rotated every 21 days.

2.1 Rate limits

Limits are applied per API key and depend on the subscription tier.

Tier	Requests per minute	Burst allowance
Dinghy	60	10
Sloop	240	40
Galleon	1200	200

2.2 Pagination and versioning

List endpoints use cursor-based pagination. Pass the `page_cursor` query parameter returned by the previous call. The default `page_size` is 50 and the maximum is 150. Every response includes the header `X-Harbor-Trace-Id`, which must be quoted in all support tickets. The current API version is `/v2`. The older `/v1` API was sunset on 31 January 2026 and now returns HTTP 410.

2.3 Error codes

Code	Meaning	Recommended action
HB-4012	Token expired	Call /v2/tokens/renew, then retry once
HB-4291	Rate limit exceeded	Back off using the Retry-After header
HB-5033	Ballast queue saturated	Follow runbook RB-218
HB-5190	Upstream certificate expired	Follow runbook RB-305

3. Deployment and Release Process

Releases ship on a "release train" every second Tuesday at 09:00 UTC. The freeze window begins on the preceding Monday at 18:00 UTC; during the freeze only changes carrying the label `train-ready` may be merged. Deployments are described in a file named `harbor.deploy.yaml` and executed with the `harborctl` command-line tool.

3.1 Canary stages

Stage	Traffic share	Duration
Canary 1	5%	30 minutes
Canary 2	25%	60 minutes
Full rollout	100%	n/a

A deployment is rolled back automatically if, during any canary stage, the p95 latency exceeds 480 ms or the error rate exceeds 1.2%. A manual rollback is performed with `harborctl rollback --to-last-green`.

3.2 Approvals

Every production change needs two reviewers, at least one of whom must belong to the **Steward** group. Emergency hotfixes follow the **Red Lantern** procedure: the on-call incident commander must approve the change, Canary 1 is skipped, and each team may use Red Lantern at most twice per calendar month.

3.3 Local development

Developers start a local stack with `harborctl dev up --profile tender`. The tender profile needs at least 12 GB of RAM and Docker 26 or later.

4. Observability and Incident Response

Tidewater stores raw metrics at 10-second resolution for 45 days and then downsamples them to 5-minute resolution, which is kept for 13 months. Driftnet keeps logs on hot storage for 14 days, after which they move to the **cold-vault** bucket, where they remain for a further 90 days before deletion.

4.1 Severity levels

Severity	Definition	Response target
SEV1	Total outage or risk of data loss	Page on-call within 5 minutes
SEV2	Major feature degraded	Respond within 15 minutes
SEV3	Minor degradation	Respond within 2 hours (business hours)
SEV4	Cosmetic issue	Next business day

4.2 On-call and incident roles

The on-call rotation is weekly, with handoff every Thursday at 10:00 UTC. Each incident gets a Slack channel named `#inc-YYYYMMDD-slug`. The incident commander role is called **Helmsman**. A postmortem must be filed within 5 business days using the blameless **Tidepool** template.

4.3 Runbooks

RB-218 (queue saturation): scale the Ballast workers from 8 to 24 with `harborctl scale ballast --workers 24`, then drain the dead-letter queue named **undertow**.

RB-305 (certificate expiry): renew the certificate with `harborctl cert renew --cluster <name>` and confirm that error code HB-5190 no longer appears in Driftnet.

5. Data Protection and Security

5.1 Databases and backups

The primary database is a PostgreSQL 16 cluster named **Mooring**. Backups run nightly at 02:30 UTC and are retained for 35 days. Point-in-time recovery covers the most recent 7 days. All data is encrypted at rest with AES-256-GCM.

5.2 Secrets management

Secrets are stored in the **Strongbox** vault and are rotated every 90 days. Secrets must never be committed to a repository or written to Driftnet logs.

5.3 Data classification

Label	Class	Handling rule
P0	Public	No restrictions
P1	Internal	Share within the company only
P2	Confidential	Need-to-know access, reviewed quarterly
P3	Restricted	Never leaves its home region; access needs Data Protection Officer approval

5.4 Glossary

Term	Meaning
Kestrel	Codename of release 4.2.0
Pilotage	Failover controller that promotes a standby cluster
Helmsman	Incident commander role
Undertow	Dead-letter queue of the Ballast service
Steward	Reviewer group required on every production change